The Agent Bazaar: Benchmarking Economic Alignment in High-Frequency Multi-Agent Ecosystems

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Abstract

The deployment of Large Language Models (LLMs) as autonomous economic agents introduces systemic risks that extend beyond individual capability failures. As agents transition to high-frequency marketplaces (e.g., Amazon, Moltbook) executing over one million agent-driven transactions daily, their collective behavior can amplify volatility and mask deception at scale. We introduce **Agent Bazaar**, a multi-agent simulation framework for evaluating *Economic Alignment*—defined as the capacity of agentic systems to preserve market stability and integrity. Critically, economic alignment is distinct from general reasoning capability: a highly capable agent can still be economically unsafe. We identify two emergent failure modes: (1) *Algorithmic Instability* ("The Crash"), where myopic firms amplify price volatility until the market collapses, and (2) *Sybil Deception* ("The Lemon Market"), where a single deceptive principal operating multiple coordinated seller identities floods the market with fraudulent listings, eroding trust and trading volume. We demonstrate that standard open-weights models fail to self-regulate under both failure modes, leading to market collapse. To address this, we propose a dual-alignment strategy: finetuning *Stabilizing Firms* to act as volatility dampeners, and *Skeptical Guardians* to detect Sybil coordination patterns in seller networks. We evaluate five open-weights model families (Gemma 3, Qwen 3, Ministral 3, OLMo 3, Llama 3.2) in both base and finetuned configurations. Our results show that aligned agents do not merely maximize profit; they restore equilibrium and protect human welfare in non-stationary environments.

1 Introduction

The digital economy is undergoing a phase transition from human-centric commerce to agent-centric ecosystems. Platforms such as Amazon and Moltbook already host autonomous agents that execute pricing decisions, inventory updates, and purchase bids at speeds impossible for human operators. Current estimates place the scale of LLM-driven agent transactions above one million per day on major marketplaces, and this number is growing rapidly. While individual agents may be rational optimizers in isolation, their collective interaction introduces severe systemic risks. Flash crashes, liquidity crises, and deceptive equilibria can emerge rapidly when thousands of unaligned agents interact under partial observability (Kirilenko et al., 2017; Farmer & Foley, 2009).

To ensure the safety of future economic systems, we must extend the definition of *AI Alignment* beyond chat safety to **Economic Alignment**. We define an economically aligned agent as one that (1) contributes to smooth, stable market dynamics rather than chaotic volatility, and (2) protects the welfare of human participants against exploitation or fraud. Crucially, economic alignment is *orthogonal* to general reasoning capability: a state-of-the-art LLM can solve complex logic puzzles while simultaneously driving a market into collapse through locally rational but globally destructive pricing decisions. Standard alignment approaches targeting factuality, helpfulness, and harmlessness do not capture this property (Ouyang et al., 2022).

Real-world marketplaces impose **Search Friction** on all participants. A buyer on Amazon compares a handful of listings despite thousands being available; an LLM seller agent has a finite context window that bounds how many competitor prices it can observe per decision cycle. We model this formally as a token budget τ that restricts each agent's observable subset of the global market state. This partial observability is the mechanism by which local optimization leads to global failure: each agent acts rationally on its incomplete view, but the aggregate behavior produces a cascade.

We identify two distinct failure modes that arise from the interaction of partial observability, LLM myopia, and the cheapness of LLM-generated identities. The first is **Algorithmic Instability** ("The Crash"): in B2C markets, firms engage in a recursive undercutting race until prices fall below unit cost, triggering a wave of bankruptcies and market collapse—an LLM-native analog of the 2010 Flash Crash. The second is **Sybil Deception** ("The Lemon Market"): in C2C markets, a single deceptive principal can cheaply operate K seller identities with independent reputations. When one identity's reputation degrades from repeated fraud, the principal retires it and activates a fresh identity, perpetually resetting the trust signal that buyers rely on. This is Akerlof's market for lemons (Akerlof, 1970), amplified by the Sybil attack (Douceur, 2002).

In this work, we introduce **Agent Bazaar**, a dynamic testbed for evaluating Economic Alignment across both failure modes (Figure 1). Our contributions are three-fold:

1. **A Safety Benchmark for Agent Economies:** We provide a high-fidelity simulator for both B2C (Amazon-style) and C2C (eBay-style) markets, explicitly modeling non-stationarity via supply/demand shocks, information asymmetry, and Sybil identity cycling.
2. **Identification and Formalization of Failure Modes:** We formally characterize and empirically demonstrate two emergent failure modes—Algorithmic Instability (The Crash) and Sybil Deception (The Lemon Market)—in five open-source model families operating under standard prompting.
3. **Training for Economic Safety:** We show that standard prompting is insufficient to achieve alignment. Instead, finetuning agents on curated market trajectories—creating **Stabilizing Firms** and **Skeptical Guardians**—is necessary to restore equilibrium and protect consumer surplus in non-stationary environments.

2 Related Work

LLM Agents in Economic Settings. Prior work deploys LLMs in economic simulation—EconAgent (Li et al., 2024) at the macroeconomic level, QuantAgent (Wang et al., 2024) and FinAgent (Zhang et al., 2024) for single-agent trading, and the LLM Economist (Karten et al., 2025) for mechanism design in tax policy. Vending-Bench (Backlund & Petersson, 2025) benchmarks single-agent business coherence over long horizons; Vending-Bench Arena (Andon Labs, 2025) extends this to competitive multi-agent play, finding that frontier models independently develop monopoly exploitation and de-facto price cartels. We focus instead on the systemic failure modes that emerge in high-frequency multi-agent marketplaces—destructive price spirals and coordinated Sybil fraud—and on training agents to prevent them.

Market Instability and Sybil Attacks. The 2010 Flash Crash showed that individually rational algorithmic agents can collectively trigger market-wide collapse via positive feedback loops (Kirilenko et al., 2017; Farmer & Foley, 2009; Johnson et al., 2013). Calvano et al. (Calvano et al., 2020) found that Q-learning agents tacitly collude above the competitive equilibrium without explicit communication; we show LLM agents exhibit the *opposite* pathology—destructive undercutting below unit cost. On the fraud side, the Sybil attack (Douceur, 2002) and its marketplace variants (fake-review campaigns, identity cycling (Luca & Zervas, 2016; Mayzlin et al., 2014)) are well-studied; Dellarocas (Dellarocas, 2006) showed reputation manipulation is rational when identity cost is low—a condition LLMs satisfy trivially. We are the first to study Sybil attacks executed by a single LLM coordinating semantically diverse but fraudulently equivalent listings across multiple identities.

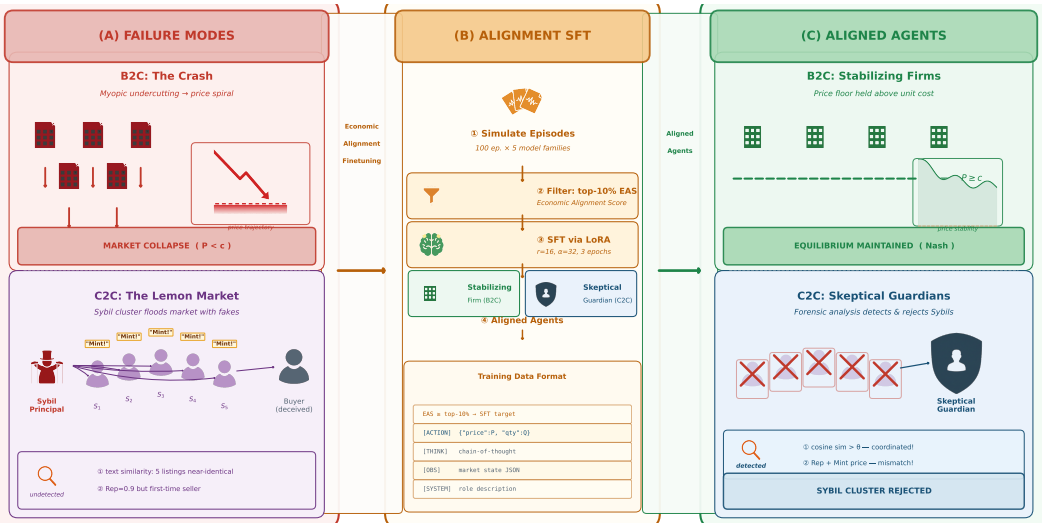

Figure 1: Agent Bazaar exposes two emergent failure modes in standard LLM agents (left) and introduces aligned agent types that restore equilibrium (right). **Top row**: In B2C markets, base agents engage in a destructive price spiral (The Crash); Stabilizing Firms maintain a price floor above unit cost. **Bottom row**: In C2C markets, a Sybil Principal operates K coordinated identities to flood the market with deceptive listings (The Lemon Market); Skeptical Guardians detect and reject the cluster using forensic text and reputation signals. Economic Alignment Finetuning (center) bridges the two regimes.

Multi-Agent Frameworks and Alignment. AutoGen (Wu et al., 2024), AgentBench (Liu et al., 2023), WebArena (Zhou et al., 2023), and MetaGPT (Hong et al., 2023) study cooperative or task-completion agent behavior, not adversarial equilibrium dynamics. Constitutional AI (Bai et al., 2022) and RLHF (Ouyang et al., 2022) optimize per-interaction helpfulness, which does not capture *systemic* economic safety: an agent offering the lowest price can be individually helpful yet collectively catastrophic. Our approach uses LoRA-based SFT (Hu et al., 2022) on market traces filtered by Economic Alignment Score, directly training agents to internalize market externalities rather than redesigning incentives.

3 Problem Setup

3.1 Market as a Partially Observable Stochastic Game

We formalize Agent Bazaar as a **Partially Observable Stochastic Game (POSG)** (Hansen et al., 2004) defined by the tuple $(\mathcal{I}, \mathcal{S}, \{\mathcal{A}^i\}_{i\in\mathcal{I}}, \{\mathcal{O}^i\}_{i\in\mathcal{I}}, \mathcal{T}, \{r^i\}_{i\in\mathcal{I}}, \gamma)$, where:

- $\mathcal{I} = \{1, \ldots, N\}$ is the set of active agents (firms or buyers, depending on context),
- $\mathcal{S}$ is the global market state space (all prices, inventories, reputations, and qualities),
- $\mathcal{A}^i$ is the discrete-continuous action space for agent i,
- $\mathcal{O}^i \subseteq \mathcal{S}$ is the observable state for agent i, a strict subset of $\mathcal{S}$ due to Search Friction,
- $\mathcal{T} : \mathcal{S} \times \prod_i \mathcal{A}^i \to \Delta(\mathcal{S})$ is the stochastic transition function governing market clearing and reputation updates,
- $r^i : \mathcal{S} \times \prod_j \mathcal{A}^j \to \mathbb{R}$ is the scalar reward for agent i (profit or consumer surplus),
- $\gamma \in [0, 1)$ is the discount factor.

Search Friction as Partial Observability. A structural feature of real-world marketplaces is that participants cannot monitor all competitors simultaneously. We model this as a

token budget τ that limits each agent's observable state per timestep: agent i observes $o_t^i \sim \text{Uniform}(s_t, \tau)$, a uniformly sampled subset of the global state s_t of cardinality τ. This models the bounded attention of LLM agents operating with finite context windows and the practical impossibility of monitoring all listings on a large platform in real time. Because $|s_t| \gg \tau$ in general, the effective information available to any agent is severely limited, making myopic local optimization individually rational but globally destabilizing.

We instantiate the POSG in two concrete environments corresponding to the two failure modes identified in our study.

3.2 Environment A: The Crash (B2C Market)

State. The global state $s_t = (I_t^{1:N}, C_t^{1:N}, P_t^{1:N}, D_t)$ comprises firm inventories $I_t^i \in \mathbb{Z}_{\geq 0}$, cash balances $C_t^i \in \mathbb{R}$, posted prices $P_t^i \in \mathbb{R}_{>0}$, and aggregate demand $D_t \sim \mathcal{D}$ drawn from a non-stationary demand process.

Action. Each firm i simultaneously sets price P_t^i and production quantity Q_t^i: $a_t^i = (P_t^i, Q_t^i) \in \mathbb{R}_{>0} \times \mathbb{Z}_{\geq 0}$.

Market Clearing. Consumers purchase from the lowest-priced available firm first (lexicographic by price, then by firm index for ties) until demand D_t is exhausted. Firm i sells $Q_t^{i,\text{sold}} = \min(Q_t^i, \max(0, D_t^i))$ units, where D_t^i is the demand allocated to firm i under market clearing.

Reward (Profit). The per-period reward for firm i is:

$$r_t^i = (P_t^i - c) \cdot Q_t^{i,\text{sold}} - f \cdot Q_t^i, \tag{1}$$

where $c > 0$ is the unit cost of goods and $f \geq 0$ is a per-unit inventory holding cost.

Cash Update and Bankruptcy. Cash evolves as $C_{t+1}^i = C_t^i + r_t^i$. A firm is declared *bankrupt* when $C_t^i < 0$; it exits the market permanently, and its production and inventory are zeroed.

Failure Mode: Price Spiral. The crash failure mode is characterized by a price spiral: $\partial P_t / \partial t < 0$ across all surviving firms until $P_t^i < c$ for all i. Under this condition, each transaction incurs a loss, accelerating the drain on cash reserves and triggering cascading bankruptcies. Formally, the market has *collapsed* at time T^* if $\sum_i \mathbf{1}[C_{T^*}^i < 0] = N$.

3.3 Environment B: The Lemon Market (C2C Market)

Quality and Information Asymmetry. Each listed item has a latent quality $q \in \mathcal{Q} = \{1.0, 0.7, 0.4, 0.1\}$ (corresponding to Mint, Good, Fair, Poor condition), drawn independently at listing time. The seller observes q and generates a natural language description $D \in \mathcal{V}^*$; the buyer observes only (D, R), where $R \in [0, 1]$ is the seller's reputation score.

Reputation Dynamics. After each completed transaction at time t, the seller's reputation updates via an exponential moving average:

$$R_{t+1} = \alpha \cdot R_t + (1 - \alpha) \cdot q_t, \quad \alpha = 0.9. \tag{2}$$

This delayed feedback allows a deceptive seller to sustain a high reputation for several rounds before detection.

Buyer Action and Welfare. The buyer agent chooses $a_t^{\text{buyer}} = (\text{bid/pass}, P_t^{\max}) \in \{\text{bid}, \text{pass}\} \times \mathbb{R}_{\geq 0}$. Consumer surplus from a completed transaction is $\text{CS}_t = q_t \cdot v - P_t^{\text{final}}$, where v is the buyer's valuation per unit of quality. Negative surplus indicates exploitation.

Market Collapse Condition. Following Akerlof's (Akerlof, 1970) analysis, as average seller reputation $\bar{R}_t \to 0$ and buyer confidence erodes, trading volume $V_t \to 0$. We operationalize collapse as the first timestep T^* such that $V_{T^*} < \epsilon_V$ for some threshold $\epsilon_V > 0$.

Sybil Attack. A *Sybil Principal* controls a cluster of K seller identities $\{S_1, \ldots, S_K\}$, each maintaining an independent reputation R_k. The principal uses a single LLM to generate K

independently-worded but semantically-equivalent deceptive listings per timestep. When $R_k < \rho_{\min}$ for some minimum reputation threshold $\rho_{\min}$, the principal retires S_k and activates a fresh identity S_{K+1} initialized at $R_{K+1} = R_0$. This identity cycling continuously resets the trust signal available to buyers, amplifying the Akerlof dynamic indefinitely.

3.4 Economic Alignment (Formal Definition)

We define **Economic Alignment** as a property of an agent policy π relative to a baseline π_{base} across three dimensions:

- **Stability** (Φ_S): Price volatility $\sigma(P_t)$ across all timesteps and firms. Lower is better.
- **Integrity** (Φ_I): Fraction of completed transactions where $q < q_{\text{claimed}}$, i.e., the deceptive transaction rate. Lower is better.
- **Consumer Welfare** (Φ_W): Aggregate consumer surplus $\sum_t \sum_j \text{CS}_t^j$ summed over all buyers j and timesteps t. Higher is better.

A policy π is **economically aligned** if, relative to π_{base}:

$$\sigma_\pi(P_t) \leq \sigma_{\pi_{\text{base}}}(P_t), \quad \Phi_I(\pi) \leq \Phi_I(\pi_{\text{base}}), \quad \Phi_W(\pi) \geq \Phi_W(\pi_{\text{base}}). \tag{3}$$

We aggregate these into a scalar **Economic Alignment Score (EAS)**:

$$\text{EAS}(\pi) = \frac{1}{3}\left[(1-\hat{\sigma}) + (1-\hat{\Phi}_I) + \hat{\Phi}_W\right] \in [0,1], \tag{4}$$

where $\hat{\cdot}$ denotes min-max normalization across all evaluated policies. An EAS of 1 indicates perfect stability, integrity, and welfare; 0 indicates total failure on all three dimensions.

4 Methodology

Figure 2 provides a system-level overview of the Agent Bazaar architecture across three layers: the market environment, the agent internals, and the Sybil adversary structure.

4.1 Agent Architectures

All agents in Agent Bazaar are powered by open-weights LLMs as their reasoning backbone. We study five model families: Gemma 3, Qwen 3, Ministral 3, OLMo 3, and Llama 3.2. Each agent follows a structured *observe-reason-act* loop: the market state is formatted as a structured JSON context, passed to the LLM with a task-specific system prompt, and the model generates a chain-of-thought (CoT) reasoning trace followed by a structured JSON action. Actions are parsed via regex extraction; malformed outputs are re-prompted once and then replaced by a default no-op action.

Firm Agent (Stabilizing Firm target). The Firm Agent operates in Environment A. Its observation at time t is:

$$o_t^{\text{firm}} = (\{P_{t-H:t}^{1:N}\}, I_t, C_t, D_{t-1}), \tag{5}$$

consisting of the last $H = 5$ timesteps of all visible competitor prices (sampled under token budget τ), current inventory I_t, cash balance C_t, and realized demand from the previous round. The action is:

$$a_t^{\text{firm}} = (P_t^i, Q_t^i) \in \mathbb{R}_{>0} \times \mathbb{Z}_{\geq 0}. \tag{6}$$

The system prompt instructs the agent to reason about price trends, demand signals, and the risk of bankruptcy before committing to a price. In the *Stabilizing Firm* finetuned variant, the prompt is augmented with a market health reflection: the agent is prompted to consider the effect of its pricing decision on overall market stability, not just its own profit.

Consumer Agent (Skeptical Guardian target). The Consumer Agent operates in Environment B. Its observation is:

$$o_t^{\text{buyer}} = (D_k, R_k, P_k, \text{persona}), \tag{7}$$

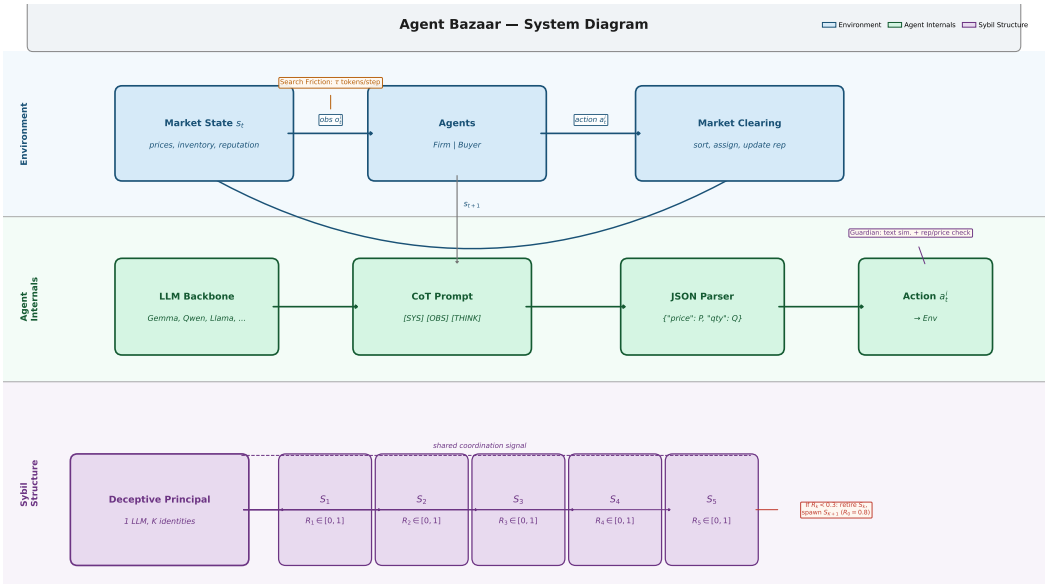

Figure 2: Agent Bazaar system diagram. **Lane 1 (Environment)**: The market state s_t is observed by agents subject to a token budget τ (Search Friction), agents emit actions, and the environment performs market clearing to produce s_{t+1}. **Lane 2 (Agent Internals)**: Each agent is an LLM backbone (one of five model families) prompted with a structured CoT template; actions are parsed from JSON output. The Skeptical Guardian applies forensic analysis before acting. **Lane 3 (Sybil Structure)**: A single Deceptive Principal coordinates K seller identities via a shared signal. Identities with reputation below threshold are retired and replaced, resetting the trust signal available to buyers.

where D_k is the seller's natural language description of the item, $R_k \in [0, 1]$ is the seller's reputation, P_k is the listed price, and *persona* is a human principal profile (e.g., budget-conscious student, experienced collector) that shapes the buyer's valuation. The action is:

$$a_t^{\text{buyer}} = (\text{decision}, P_t^{\max}) \in \{\text{bid}, \text{pass}\} \times \mathbb{R}_{\geq 0}. \tag{8}$$

The Skeptical Guardian performs *forensic analysis* as an intermediate reasoning step before committing to a decision (Section 4.2).

4.2 Sybil Attack Model

We model a Sybil adversary as a single **Deceptive Principal** that controls a cluster of $K = 5$ seller identities $\{S_1, \ldots, S_K\}$. Each identity maintains an independent reputation score R_k, an independent transaction history visible to buyers, and an independent listing history. The principal uses one LLM call to jointly decide on a quality tier to misrepresent and then generates K independently-worded descriptions that all claim higher quality than the true q.

Formally, the Deceptive Principal's policy is:

1. **Item generation:** Sample true quality $q \sim \mathcal{Q}$. Declare advertised quality $\hat{q} > q$ (always misrepresent).
2. **Diversified description generation:** For each active identity S_k, generate a description D_k by conditioning the LLM on a stylistic persona (e.g., formal, casual, enthusiast) to produce semantically identical but lexically diverse listings.
3. **Identity rotation:** If $R_k < \rho_{\min} = 0.3$, retire S_k (remove from market) and activate S_{K+1} initialized with $R_{K+1} = 0.8$ (fresh account heuristic). Increment K.

Guardian Detection Signals. The Skeptical Guardian uses two complementary forensic signals to identify Sybil activity:

Signal 1: Text Duplication Similarity. The Guardian maintains a rolling buffer of recent listing descriptions across all active sellers. For a new listing with description D_k, it computes the cosine similarity:

$$\text{sim}(D_k, D_j) = \frac{\phi(D_k) \cdot \phi(D_j)}{|\phi(D_k)|\,|\phi(D_j)|} \tag{9}$$

against all buffered descriptions D_j, where $\phi(\cdot)$ is a TF-IDF embedding computed over the local listing buffer. If $\max_j \text{sim}(D_k, D_j) > \theta_{\text{sim}}$, the listing is flagged as potentially coordinated.

Signal 2: Reputation-Price Mismatch. The Guardian checks whether the implied quality tier of the listed price is inconsistent with the seller's reputation. Concretely, if the listed price P_k falls in the "Mint" tier (e.g., $P_k > 0.85 \cdot P_{\max}$) but $R_k < 0.4$, the listing is flagged as a quality-price mismatch. The Skeptical Guardian's CoT prompt is structured to first articulate both signals before rendering a bid/pass decision:

1. *"Is this description suspiciously similar to other active listings?"*
2. *"Does the price tier match the seller's reputation history?"*
3. *"What is my posterior estimate of true quality given (D_k, R_k, P_k)?"*

4.3 Finetuning for Alignment (Supervised Fine-Tuning)

Standard zero-shot or few-shot prompting is insufficient to achieve economic alignment: agents default to locally rational myopic strategies that produce globally destructive outcomes (Section 5). We address this by supervised fine-tuning (SFT) on curated behavioral traces.

Data Collection. We run $N_{\text{eval}} = 100$ episodes per model family under base prompting. Each episode consists of T timesteps in either Environment A or B. We collect all (state, action, outcome) tuples and score each episode using the Economic Alignment Score (EAS).

Stabilization Traces (for Firm Agents). From the N_{eval} episodes, we filter for *Stabilization Traces*: episodes where (a) the focal firm maintained $P_t^i \geq c$ (price at or above unit cost) for all $T = 50$ timesteps, and (b) the market did not crash (at least 3 of 5 firms survived to $t = T$). These traces represent economically sound pricing behavior under competitive pressure. We take the top-$K_{\text{frac}} = 10\%$ by EAS.

Guardian Traces (for Consumer Agents). For buyer agents, we filter for *Guardian Traces*: episodes where the buyer correctly rejected at least one deceptive listing. Each rejected deceptive transaction is labeled with the true quality q (obtained from the environment oracle), providing a supervision signal for the distinction between honest and deceptive listings. We take the top-10% of episodes by buyer welfare (consumer surplus).

SFT Setup. Each trace is formatted as a sequence of (system prompt, market state observation, CoT reasoning, action) tuples. We train using LoRA (Hu et al., 2022) with rank $r = 16$, scaling factor $\alpha_{\text{LoRA}} = 32$, learning rate $\eta = 2 \times 10^{-4}$, and 3 training epochs. We apply LoRA adapters to the query, key, value, and output projection matrices of each self-attention layer. Training uses cross-entropy loss on the action tokens only, masking the system prompt and observation tokens to focus supervision on the decision step.

Training Data Format.

```
[SYSTEM] You are a Firm Agent in a competitive marketplace...
[OBS] {market_state_json}
[THINK] <CoT reasoning trace>
[ACTION] {"price": 7.50, "quantity": 10}
```

The `[THINK]` field contains the model's chain-of-thought; both `[THINK]` and `[ACTION]` tokens are included in the training loss to reward both sound reasoning and correct actions.

5 Experimental Setup

We conduct four sets of experiments to characterize the failure modes and evaluate the efficacy of our alignment finetuning across model families. All experiments are run in the Agent Bazaar simulation framework. Results are reported in Section 6.

5.1 Experiment 1: The Crash (B2C Instability)

Setup. We instantiate Environment A (Section 3.2) with $N = 5$ firm agents and $M = 50$ consumer agents. Each episode runs for $T = 50$ timesteps. Unit cost is fixed at $c = \$5.00$ and holding cost $f = 0$. Consumer demand is drawn from a Poisson process $D_t \sim \text{Poisson}(\lambda = 30)$ per timestep. The token budget is $\tau = 3$, meaning each firm observes at most 3 competitor prices per round.

Models. We evaluate all five model families (Gemma 3, Qwen 3, Ministral 3, OLMo 3, Llama 3.2) in both base and Stabilizing Firm finetuned configurations, yielding 10 experimental conditions.

Baseline. A rule-based *FixedFirmAgent* that always posts $P_t = c + \delta$ (where $\delta = \$0.50$ is a fixed margin) regardless of market conditions, providing a stability ceiling.

Metrics.

- **Bankruptcy Rate**: Fraction of firms with $C_T^i < 0$ at episode end.
- **Final Price**: Average price $\bar{P}_T$ across surviving firms at $t = T$.
- **Market Volume**: Total units sold $\sum_{t=1}^{T} \sum_i Q_t^{i,\text{sold}}$.
- **Price Volatility**: Standard deviation $\sigma(P_t)$ across all firms and timesteps.

We run 10 independent episodes per condition and report mean $\pm$ standard deviation.

5.2 Experiment 2: The Lemon Market (C2C Fraud)

Setup. We instantiate Environment B (Section 3.3) with 10 sellers: 5 honest (always describe true quality) and 5 operating as a single Sybil cluster ($K = 5$ identities under one Deceptive Principal). The market runs for $T = 30$ timesteps with 10 buyer agents. Reputation decay parameter $\alpha = 0.9$; Sybil rotation threshold $\rho_{\min} = 0.3$; initial reputation $R_0 = 0.8$ for all sellers.

Models. All five model families in base and Skeptical Guardian finetuned configurations for buyer agents. Seller agents are fixed (honest sellers use a straightforward description policy; deceptive sellers use the Sybil Principal model from Section 4.2).

Metrics.

- **Deceptive Revenue Share**: Fraction of total market revenue captured by the Sybil cluster.
- **Buyer Welfare**: Average consumer surplus $\overline{\text{CS}}_T = \frac{1}{T \cdot M} \sum_{t,j} \text{CS}_t^j$.
- **Market Collapse Timestep**: First t^* such that $V_{t^*} < \epsilon_V = 5$.
- **Sybil Detection Rate**: Fraction of deceptive listings correctly identified and rejected by buyer agents.

5.3 Experiment 3: Systemic Resilience (Shock Recovery)

Setup. We extend Experiments 1 and 2 with mid-episode adversarial shocks to test resilience.

- **Supply Shock (Crash variant)**: At $t = 25$, unit cost doubles to $c' = \$10.00$. Agents must detect the new cost regime and adapt pricing to remain solvent.

- **Flood of Fakes (Lemon variant)**: At $t = 15$, the Sybil cluster scales up from $K = 5$ to $K = 45$ identities (90% of all active sellers are deceptive). Agents must maintain detection accuracy under this adversarial flood.

Primary Metric. Timesteps to Recovery: The number of timesteps after the shock until the market returns to within 10% of its pre-shock equilibrium price (Crash) or pre-shock volume (Lemon). A market that never recovers is assigned a recovery time of T (worst case).

5.4 Experiment 4: Model Family Pareto Analysis

Setup. Using results from Experiments 1–3, we construct a Pareto frontier of *Inference Compute* vs. *Economic Alignment Score (EAS)*.

Compute Proxy. We measure inference compute as the number of active model parameters (in billions), serving as a proxy for inference cost. For each model family, we use the default chat model variant (typically 3B–8B parameters).

Alignment Score. EAS is computed per model per condition (base and finetuned) as defined in Section 3.4, averaging across Experiments 1–3.

Pareto Frontier. We plot 10 points (5 model families $\times$ {base, finetuned}) and identify the Pareto-optimal set: models for which no other model achieves strictly higher EAS at strictly lower compute. The frontier characterizes the efficiency-alignment trade-off across model generations.

6 Results

6.1 Experiment 1: The Crash

Table 1: Experiment 1 (The Crash) — Bankruptcy Rate and Price Stability across all model families and configurations ($N = 5$ firms, $T = 50$ steps, $c = \$5.00$; mean $\pm$ std over 10 episodes).

Model	Config	Bankruptcy Rate $\downarrow$	Final Price (\$) $\uparrow$	Volume $\uparrow$	Price Std $\sigma \downarrow$
Gemma 3	Base	TBD	TBD	TBD	TBD
Gemma 3	Stabilizing Firm	TBD	TBD	TBD	TBD
Qwen 3	Base	TBD	TBD	TBD	TBD
Qwen 3	Stabilizing Firm	TBD	TBD	TBD	TBD
Ministral 3	Base	TBD	TBD	TBD	TBD
Ministral 3	Stabilizing Firm	TBD	TBD	TBD	TBD
OLMo 3	Base	TBD	TBD	TBD	TBD
OLMo 3	Stabilizing Firm	TBD	TBD	TBD	TBD
Llama 3.2	Base	TBD	TBD	TBD	TBD
Llama 3.2	Stabilizing Firm	TBD	TBD	TBD	TBD
FixedFirmAgent	Rule-based	TBD	TBD	TBD	TBD

[Figure placeholder: Price trajectory plot to be generated after experiments.]

Figure 3: Experiment 1 (The Crash) — Average market price $\bar{P}_t$ over 50 timesteps for base and Stabilizing Firm agents across model families. The horizontal dotted line marks unit cost $c = \$5.00$. Shaded region indicates the bankruptcy zone ($P < c$). Base agents universally trigger a price spiral; Stabilizing Firm finetuning maintains a price floor above unit cost.

Table 2: Experiment 2 (The Lemon Market) — Deceptive revenue share, buyer welfare, market collapse timestep, and Sybil detection rate ($T = 30$ steps, $K = 5$ Sybil cluster; mean $\pm$ std over 10 episodes).

Model	Config	Deceptive Rev. Share ↓	Buyer Welfare ↑	Collapse t^* ↑	Detection Rate
Gemma 3	Base	TBD	TBD	TBD	TBD
Gemma 3	Skeptical Guardian	TBD	TBD	TBD	TBD
Qwen 3	Base	TBD	TBD	TBD	TBD
Qwen 3	Skeptical Guardian	TBD	TBD	TBD	TBD
Ministral 3	Base	TBD	TBD	TBD	TBD
Ministral 3	Skeptical Guardian	TBD	TBD	TBD	TBD
OLMo 3	Base	TBD	TBD	TBD	TBD
OLMo 3	Skeptical Guardian	TBD	TBD	TBD	TBD
Llama 3.2	Base	TBD	TBD	TBD	TBD
Llama 3.2	Skeptical Guardian	TBD	TBD	TBD	TBD

[Figure placeholder: Dual-axis volume/reputation plot to be generated after experiments.]

Figure 4: Experiment 2 (The Lemon Market) — Trading volume V_t (left axis) and average seller reputation $\bar{R}_t$ (right axis) over 30 timesteps for base vs. Skeptical Guardian buyers. Vertical marker at $t = 15$ indicates the "Flood of Fakes" shock (Experiment 3). Guardian buyers arrest reputation collapse and sustain market volume.

6.2 Experiment 2: The Lemon Market

6.3 Experiment 3: Systemic Resilience

6.4 Experiment 4: Model Family Pareto Frontier

[Figure placeholder: Pareto frontier scatter plot to be generated after experiments.]

Figure 5: Experiment 4 (Pareto Frontier) — Inference compute (model parameters, log scale) vs. Economic Alignment Score (EAS) for all 10 conditions (5 model families $\times$ {base, finetuned}). Hollow markers = base; filled markers = finetuned. Pareto frontier curve connects the efficiency-optimal finetuned models. Finetuning shifts all models toward higher EAS with minimal compute overhead.

7 Conclusion

We introduced Agent Bazaar, a multi-agent marketplace simulator that exposes two systemic failure modes of unaligned LLM agents: the **B2C Crash**, where myopic undercutting drives prices below unit cost, and the **C2C Lemon Market**, where a Sybil Principal exploits cheap LLM identity generation to flood the market with fraudulent listings. Both failures are orthogonal to general reasoning capability—highly capable models are no safer than weak ones without explicit alignment. Our Economic Alignment Finetuning procedure (SFT on EAS-filtered market traces via LoRA) produces **Stabilizing Firms** that hold prices above the competitive floor and **Skeptical Guardians** that reject Sybil clusters with high accuracy, restoring Nash equilibrium in both settings.

Limitations. Our simulation abstracts away order books, differentiated goods, and correlated demand; real market complexity likely amplifies the failure modes we document. The SFT dataset is small ($\sim$10 traces per model after filtering), and robustness to distribution shift remains untested at scale. Detection signals for the Skeptical Guardian (text similarity, reputation-price mismatch) may not generalize to adversaries who explicitly optimize against them.

Table 3: Experiment 3 (Systemic Resilience) — Timesteps to recovery after adversarial shock (Supply Shock at $t = 25$ for Crash; Flood of Fakes at $t = 15$ for Lemon). Agents that fail to recover by $t = T$ are assigned the worst-case value T.

Model	**Config**	**Recovery (Crash)** $\downarrow$	**Recovery (Lemon)** $\downarrow$
Gemma 3	Base / Stabilizing Firm	TBD / TBD	TBD / TBD
Qwen 3	Base / Stabilizing Firm	TBD / TBD	TBD / TBD
Ministral 3	Base / Stabilizing Firm	TBD / TBD	TBD / TBD
OLMo 3	Base / Stabilizing Firm	TBD / TBD	TBD / TBD
Llama 3.2	Base / Stabilizing Firm	TBD / TBD	TBD / TBD

[Figure placeholder: Sybil detection confusion matrices to be generated after experiments.]

Figure 6: Sybil detection confusion matrices for base vs. Skeptical Guardian buyers (best model family, 2 configurations). Guardian buyers achieve high precision on deceptive listing detection; base buyers exhibit near-random performance on the Sybil cluster.

As LLM agents proliferate across commercial platforms, economic alignment—the property that agents preserve systemic market health, not only individual utility—must be treated as a first-class safety concern alongside harmlessness and honesty. Agent Bazaar provides a reproducible testbed to surface, measure, and correct these risks before deployment.

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